

TASK 9

Calculate the message digest of a text using the SHA-1 algorithm in JAVA.

Aim: To calculate the message digest of a text using the SHA-1 algorithm in JAVA.

Program:

```
package com.islab;
import java.security.*;
public class SHA1
{
public static void main(String[] a)
{
try
{
    MessageDigest md = MessageDigest.getInstance("SHA1");
    System.out.println("Message digest object info: ");
    System.out.println(" Algorithm = " + md.getAlgorithm());
    //System.out.println(" Provider = " + md.getProvider());
    System.out.println(" ToString = " + md.toString());
    String input = "";
    md.update(input.getBytes());
    byte[] output = md.digest();
    System.out.println();
    System.out.println("SHA1(\"" + input + "\") = " + bytesToHex(output));
    input = "abc";
    md.update(input.getBytes());
    output = md.digest();
    System.out.println();
    System.out.println("SHA1(\"" + input + "\") = " + bytesToHex(output));
    input = "abcdefghijklmnopqrstuvwxyz";
    md.update(input.getBytes());
    output = md.digest();
    System.out.println();
    System.out.println("SHA1(\"" + input + "\") = " + bytesToHex(output));
    System.out.println("");
}
}
```

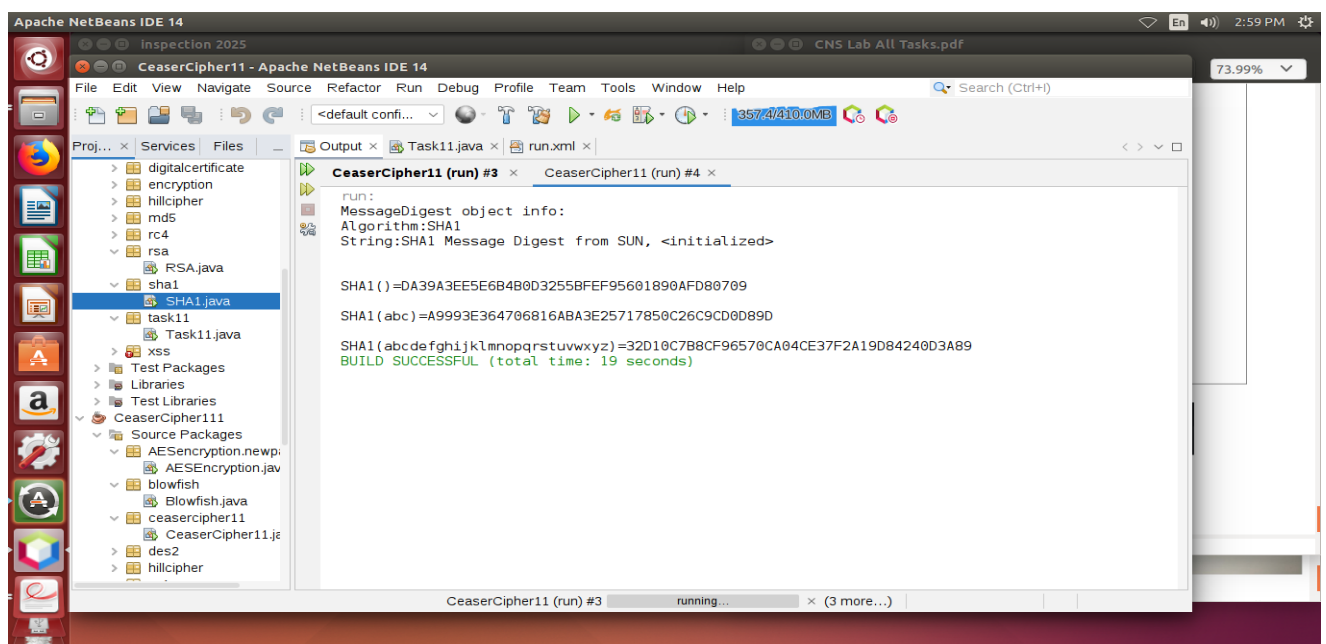
```

        catch (Exception e)
        {
            System.out.println("Exception: " + e);
        }
    }

    public static String bytesToHex(byte[] b) {
        char hexDigit[] = { '0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'A', 'B', 'C', 'D', 'E', 'F' };
        StringBuffer buf = new StringBuffer();
        for (int j = 0; j < b.length; j++) {
            buf.append(hexDigit[(b[j] >> 4) & 0x0f]);
            buf.append(hexDigit[b[j] & 0x0f]);
        }
        return buf.toString();
    }
}

```

OUTPUT:



VIVA QUESTIONS

1. What is the length of the SHA-1 hash output?
2. What is the block size used in SHA-1?
3. How many rounds does SHA-1 use in its hashing process?
4. Explain the working steps of the SHA-1 algorithm